# 2025 VCE Chemistry external assessment report

This report provides sample answers or an indication of what answers may have included. Unless otherwise stated, these are not intended to be exemplary or complete responses.

The statistics in this report may be subject to rounding, resulting in a total more or less than 100 per cent.

## Section A

The table below indicates the percentage of students who chose each option. The correct answer is indicated by grey shading.

| Question | Correct answer | % A | % B | % C | % D | Comments |
|---|---|---|---|---|---|---|
| 1 | B | 5 | **92** | 2 | 1 | Biomethane can be made quickly by anaerobic digestion of plant and animal material and so is renewable.<br>Natural gas is a fossil fuel and therefore is not classified as renewable. |
| 2 | D | 18 | 35 | 13 | **34** | Energy from 65 g = (16 x 45) + (17 x 10) + (37 x 10) = 1260 kJ<br>Energy from 100 g = 1260 x 100/65 = 1938 kJ = $1.9 \times 10^6$ J<br>The most common error among the responses was unit error. |
| 3 | C | 2 | 5 | **71** | 22 | Breaking bonds requires energy and is an endothermic process.<br>Forming bonds releases energy and is an exothermic process.<br>For a reaction that is overall exothermic, breaking bonds requires less energy than what is released as the new bonds form. |
| 4 | C | 3 | 9 | **83** | 5 | Circular economy is improved when 'waste' materials can be recycled or used. Option C is the only option that describes how 'waste' materials are being managed successfully. |
| 5 | D | 2 | 3 | 2 | **93** | $CF = VIt/\Delta T = 12.0 \times 2.50 \times 150 / 6.1 = 738\ J\ {}^{o}C^{-1}$ |
| 6 | A | **56** | 9 | 29 | 6 | $6CO_{2(g)} + 6H_2O_{(l)} \rightarrow C_6H_{12}O_{6(s)} + 6O_{2(g)}$  $\Delta H = +2840$ kJ<br>$CO_{2(g)} + H_2O_{(l)} \rightarrow {}^1/_6C_6H_{12}O_{6(s)} + O_{2(g)}$ $\Delta H = +2840/6 = +473.3$ kJ |
| 7 | B | 60 | **24** | 9 | 6 | Metal ions are generated at the anode and these have to migrate through the polymer/gel electrolyte in order to balance the negative charges being formed at the cathode.<br>While the anode generates electrons which flow through the wire to the cathode, this question refers to 'through the polymer/gel electrolyte'. |

| Question | Correct answer | % A | % B | % C | % D | Comments |
|---|---|---|---|---|---|---|
| 8 | A | 29 | 7 | **53** | 11 | Circuit A generates 3.60 V<br>Circuit B generates 2.06 V<br>Circuit C generates 3.10 V<br>Circuit D generates 1.99 V |
| 9 | D | 11 | 8 | 23 | **57** | Zn(s) does not normally react with water, whereas all three of the other metals have the potential to spontaneously react with water.<br>$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$ $E^o = -0.76$ V<br>$2H_2O(l) + 2e^- \rightleftharpoons H_2(g) + 2OH^-(aq)$ $E^o = -0.83$ V |
| 10 | ABCD | | | | | Statement II is clearly a correct statement as only hydrolysis and fermentation require enzymes to occur.<br>Statement III is also correct as during hydrolysis, glucose is formed (among other possibilities).<br>Statement I caused issues and confusion. Fermentation of glucose to form ethanol and carbon dioxide involves BOTH a reduction and oxidation process. The oxidation number of carbon changes from 0 to –2 as ethanol is formed and also from 0 to +4 as carbon dioxide is formed.<br>Since there was no option for I, II and III being correct, all four options were accepted. |
| 11 | C | 4 | 4 | **80** | 11 | Fuel efficiency relates to the effectiveness of converting the fuel into its products in terms of either time or yield. This can be enhanced by increasing the reaction rate. Using porous electrodes, which increases the surface area of the electrode, is an effective method. |
| 12 | B | 19 | **62** | 13 | 5 | $n(e^-) = Q/F = 36000 / 96500 = 0.373$ mol<br>$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$<br>$n(O_2) = n(e^-)/4 = 0.373 / 4 = 0.09326$ mol<br>$m(O_2) = n \times M = 0.09326 \times 32 = 2.98$ g |
| 13 | D | 20 | 6 | 13 | **61** | Recharge $2Ni(OH)_2 + Cd(OH)_2 \rightarrow 2NiO(OH) + Cd + 2H_2O$<br>Reduction process is therefore<br>$Cd(OH)_2 + 2e^- \rightarrow Cd + 2OH^-$ |
| 14 | C | 5 | 25 | **62** | 8 | Catalysts reduce the activation energy for a reaction and therefore lower the energy required for a successful collision. Hence the proportion of particles with $E > E_a$ will increase. |
| 15 | D | 12 | 21 | 10 | **56** | $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ $K = 0.286$ $M^{-2}$<br>$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$ $K = 1 / 0.286$ $M^2$<br>$4NH_3(g) \rightleftharpoons 2N_2(g) + 6H_2(g)$ $K = (1 / 0.286\ M^2)^2 = 12.2$ $M^4$ |

| Question | Correct answer | % A | % B | % C | % D | Comments |
|---|---|---|---|---|---|---|
| 16 | C | 9 | 16 | **67** | 8 | $\Delta H$ is negative, therefore the forward reaction is exothermic.<br>A temperature increase will cause the equation to shift in the endothermic direction (reverse reaction is favoured).<br>Therefore [NO] ↑, $[O_2]$ ↑ and $[NO_2]$ ↓ with no instantaneous changes. |
| 17 | C | 10 | 34 | **49** | 7 | Energy efficiency requires less wastage of the desired energy.<br>Benefit I will be more efficient because better conduction properties means less heat energy is being produced.<br>Benefit II will be more efficient because less heat energy is lost.<br>In both cases with less heat energy being generated from the cell, the processes occurring in an EFFC meet the requirements for design for energy efficiency. |
| 18 | A | **41** | 21 | 15 | 23 | In artificial photosynthesis, water is converted into hydrogen and oxygen gas by the sole application of sunlight as the energy source.<br>In this process, water is oxidised to produce oxygen gas and at the same time water is reduced to form hydrogen gas. Option A aligns with this.<br>Natural photosynthesis produces oxygen gas and glucose.<br>When hydrogen ions gain electrons to form hydrogen gas, this can only occur at the cathode. |
| 19 | C | 19 | 21 | **34** | 25 | Students struggled with the term 'resolution' when it was used both as a numerical value and as a conceptual property of a measuring instrument. An instrument with a higher resolution can detect smaller changes in a measured quantity, allowing measurements to be recorded in finer increments.<br>• The 50 mL beaker has a resolution of 10 mL.<br>• The 50 mL measuring cylinder has a resolution of 1 mL.<br>• The 10 mL measuring cylinder has a resolution of 0.2 mL.<br>• The 3 mL pipette has a resolution of 0.5 mL.<br>Therefore the 10 mL measuring cylinder has a highest resolution compared to the 50 mL beaker, which has the lowest resolution. |
| 20 | A | **64** | 9 | 17 | 9 | The variability in the graph must be due to random error, as data points lie both above and below the line of best fit. There is no reference line, so systematic errors cannot be established.<br>The greatest source of random error will be the subjective observation of when the colour change has occurred.<br>Options B and C would produce a systematic error in the data.<br>Option D would not produce any significant random errors without the temperature being increased significantly for some experiments and decreased significantly for others. |

| Question | Correct answer | % A | % B | % C | % D | Comments |
|---|---|---|---|---|---|---|
| 21 | B | 7 | **84** | 6 | 3 | The use of renewable organic materials will always reduce the reliance on finite non-renewable sources of organics such as fossil fuels. The other three options are possible but not definite consequences in all scenarios. |
| 22 | D | 12 | 5 | 7 | **75** | $CH_3CH_2CH_2OH + HOOCCH_2CH_2CH_3 \rightarrow CH_3CH_2CH_2OOCCH_2CH_2CH_3 + H_2O$ |
| 23 | C | 11 | 17 | **63** | 9 | Hydrolytic reactions occur when water is used to break apart large molecules into smaller ones. (Hydrolytic reactions ≡ hydrolysis reactions) |
| 24 | D | 9 | 10 | 16 | **64** | 16% = 0.45 x (Yield for Step 2)<br>%Yield for Step 2 = 16% / 0.45 = 35.5% = 36% |
| 25 | C | 14 | 7 | **59** | 19 | Fractional distillation is designed to separate volatile compounds that have similar boiling points.<br>Simple distillation requries boiling points of volatile compounds to have significantly different boiling points in order to successfully separate them.<br>Solvent extraction will be unable to separate polar compounds from each other. It is most effective when compounds have distinctly different polarities. |
| 26 | C | 6 | 10 | **60** | 24 | Geranial has a molecular formula of $C_{10}H_{16}O$<br>Linalool has a molecular formula of $C_{10}H_{18}O$ |
| 27 | A | **41** | 21 | 13 | 24 | Geranial is an aldehyde and is readily oxidised by acidified dichromate and a colour change from orange to green will occur.<br>Linalool is a tertiary alcohol and therefore cannot be readily oxidised by acidified dichromate and so no colour change will occur, it will remain orange. Therefore, the use of dichromate will enable identification of these two compounds.<br>If acidified permanganate was to be used, the colour change would be from bright purple to pale pink upon reaction, not vice versa.<br>Both compounds have C=C double bonds and will readily decolourise bromine solution.<br>Neither compound contains a carboxyl group and so neither will generate carbon dioxide gas. |
| 28 | A | **45** | 28 | 16 | 11 | If impurities are present in a compound, the melting point range will be broader and lower compared with the literature value. The fact that the stem of the question states that the vanillin is often blended with significant amounts of impurities discounts option B as being a viable response. |
| 29 | B | 32 | **50** | 11 | 7 | If one mole of a triglyceride reacts with 6 moles of iodine, then each fatty acid must contain 2 carbon to carbon double bonds.<br>Only linoleic acid has two C=C bonds. |

| Question | Correct answer | % A | % B | % C | % D | Comments |
|---|---|---|---|---|---|---|
| 30 | A | **67** | 12 | 14 | 7 | Pentan-3-one is a symmetrical ketone and so will have only three carbon signals present in a $^{13}C$ NMR spectrum.<br>Being a ketone, one of these must be found in the region of 205–220 ppm on the spectrum. |

# Section B

## Question 1a.

| Marks | 0 | 1 | 2 | Average |
|---|---|---|---|---|
| % | 9 | 48 | 42 | 1.4 |

The first mark was awarded for a correctly balanced chemical equation.

$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l) \qquad \Delta H = -2220$ kJ

The second mark was awarded for an enthalpy value of $\Delta H = -2220$ kJ, which had to be accompanied by the correct states for each chemical at SLC.

The most common errors were:

- failing to supply any enthalpy value
- failing to supply the correct states that are associated with the −2220 kJ enthalpy value
- incorrectly balancing the equation.

## Question 1b.

| Marks | 0 | 1 | 2 | Average |
|---|---|---|---|---|
| % | 7 | 20 | 73 | 1.7 |

The first mark was awarded for the correct calculation of n($C_3H_8$).

$$n(C_3H_8) = \frac{m(C_3H_8)}{M(C_3H_8)} = \frac{9}{44} = 0.2045 \text{ mol}$$

The second mark was awarded for the correct energy released with units.

Energy released = 0.2045 x 2220 = 454 kJ = 4.5 x $10^2$ kJ

The most common errors were:

- failing to quote a unit or quoting an incorrect unit
- including the negative sign and making a statement that the energy released was −454 kJ.

## Question 1c.i

| Marks | 0 | 1 | 2 | Average |
|---|---|---|---|---|
| % | 10 | 22 | 68 | 1.6 |

The first mark was awarded for the correct calculation of $\Delta T$.

$\Delta T = 90.1 – 24.1 = 66.0$ °C

The second mark was awarded for the correct calculation of energy absorbed with unit.

$q = mc\Delta T = 166.0 \times 4.18 \times 66.0 = 45{,}796$ J = 45.8 kJ

## Question 1c.ii

| Marks | 0 | 1 | Average |
|---|---|---|---|
| % | 35 | 65 | 0.7 |

Energy efficiency $= \frac{45.8}{454}$ x 100 = 10%

## Question 1d.

| Marks | 0 | 1 | 2 | 3 | 4 | Average |
|---|---|---|---|---|---|---|
| % | 34 | 19 | 16 | 19 | 12 | 1.6 |

Two significant issues arose in responses to this question.

- Some responses did not show an understanding of how to convert $m^3$ to L. This conversion was given in Item 4 of the Databook. At this stage in the calculations, this led to considerable issues when students attempted to determine the limiting reagent.
- Some responses did not establish the quantity of excess reagent that remained after the reaction.

The first two marks were awarded for establishing the $n(O_2)$ present in the 0.125 $m^3$ container (125 L).

$V(O_2) = 0.210 \times 125 = 26.25$ L and then $n(O_2) = 26.25 \div 24.8 = 1.058$ mol

or

n(gas) = 125/24.8 = 5.04 mol and then $n(O_2) = 0.210 \times 5.04 = 1.058$ mol

The third mark was awarded for correct calculation of n(excess reagent).

$n(O_2)_{excess} = n(O_2)_{inital} - n(O_2)_{needed} = 1.058 - (5 \times 0.2045) = 0.0355$ mol

The fourth mark was awarded for correct mass calculation of excess reagent with units.

$m(O_2) = 0.0355 \times 32 = 1.1$ g

## Question 1e.

| Marks | 0 | 1 | 2 | Average |
|---|---|---|---|---|
| % | 50 | 42 | 8 | 0.6 |

The first mark was awarded for stating or recognising that the energy content of ethanol is less than the energy content of propane in both kJ $g^{-1}$ or kJ $mol^{-1}$.

Heat of combustion(ethanol) = 29.7 kJ $g^{-1}$ < Heat of combustion(propane) = 40.5 kJ $g^{-1}$

$\Delta H$(ethanol) = −1370 kJ < $\Delta H$(propane) = −2220 kJ

The second mark was awarded for recognising that this difference is due to the partial oxidation of ethanol, which had already occurred with the introduction of the hydroxy group to the molecule.

A common error was stating bond enthalpies as the reason for the different energy content. This could not be accepted without full calculations based on the two complete combustion equations, which is beyond the scope of the study design.

# Question 2a.

| Marks | 0 | 1 | 2 | Average |
|---|---|---|---|---|
| % | 11 | 40 | 49 | 1.4 |

The first mark was awarded for correctly identifying the oxidation numbers.

O.N.($Cr^{3+}$) = +3 and O.N.(Cr) = 0

The second mark was awarded for using these numbers to justify why it was then a reduction process.

i.e. ‘When O.N. decrease, reduction is occurring.’

The two most common errors were:

- incorrect use of ‘3+’ as an oxidation number
- incorrect terminology being used when describing the electron transfer during reduction and oxidation processes.

# Question 2b.

| Marks | 0 | 1 | 2 | Average |
|---|---|---|---|---|
| % | 66 | 28 | 6 | 0.4 |

The first mark was awarded for showing an understanding that for an effective electroplating process, either:

- the concentration of $Cr^{3+}$ in solution should be kept constant; OR
- the rate of decrease of $Cr^{3+}$ must be the same as the rate of production of $Cr^{3+}$.

The second mark was awarded for indicating how the anode and cathode play a role in this. For example, students could have justified this by either of the following, but specific reference to the anode and cathode was required.

- Stating that $Cr^{3+}$ is produced at the anode at the same rate that the $Cr^{3+}$ is used at the cathode.
- Writing the two relevant ½ equations:

(Anode) $Cr(s) \rightarrow Cr^{3+}(aq) + 3e^-$ and (Cathode) $Cr^{3+}(aq) + 3e^- \rightarrow Cr(s)$

A significant number of responses did not recognise the need for a commercial electroplating set-up to have the electrolyte with a constant concentration and were not awarded the first mark.

# Question 2c.

| Marks | 0 | 1 | 2 | 3 | Average |
|---|---|---|---|---|---|
| % | 60 | 15 | 15 | 10 | 0.8 |

The first mark was awarded for relative strength of possible oxidants.

- $H^+$ is a stronger oxidant than $Cr^{3+}$
- The ½ equation $2H^+ + 2e^- \rightarrow H_2$ is higher on the SEP table than $Cr^{3+} + 3e^- \rightarrow Cr$

The second mark was awarded for any issue that arises from too much sulfuric acid. Any of:

- $H_2$ will be produced.
- $H^+$ will react spontaneously with the Cr being produced.
- If small quantities of $H^+$ are present and this is below standard 1 M concentrations, $H^+$ reduction reaction is less likely or its position will be moved relative 0.00 V.

The third mark was awarded for the effect of the issue identified. Any of:

- Hydrogen gas produced could be flammable/dangerous.
- Chromium plating will not occur.
- Chromium plating may be lost if left in solution after the electricity is turned off.

Students found it difficult to use the correct language and often referred to the Cr as an oxidant.

A significant number of responses implied that $H_2SO_4$ was a direct source of $H_2$ when added to the solution.

Another aspect that caused confusion related to the aqueous nature of the $H_2SO_4(aq)$ being added. These responses incorrectly stated that this would introduce water into the reaction, which would cause further reactions.

# Question 2d.

| Marks | 0 | 1 | 2 | 3 | 4 | Average |
|---|---|---|---|---|---|---|
| % | 20 | 4 | 7 | 24 | 45 | 2.7 |

The first mark was awarded for the determination of the amount of chromium.

$n(Cr) = m / M = 7.80/52.0 = 0.150$ mol

The second mark was awarded for the determination of the amount of electrons.

$n(e^-) = 3 \times n(Cr) = 3 \times 0.15 = 0.450$ mol

The third mark was awarded for the determination of charge.

$Q = n(e^-) \times F = 0.450 \times 96{,}500 = 43{,}425$ C

The fourth mark was awarded for the determination of the amount of chromium.

$I = Q / t = = 43{,}425 / (2.5 \times 60 \times 60) = 4.83$ (A)

# Question 2e.

| Marks | 0 | 1 | 2 | Average |
|---|---|---|---|---|
| % | 69 | 9 | 23 | 0.6 |

Students took two varied approaches in responding to this question, from full mathematical proofs to fully worded explanations.

The first mark was awarded for showing an understanding that for the same mass of rhodium a smaller amount of electrons was needed. To effectively do this, students needed to recognise that the charge on each ion was identical, so each ion needed the same quantity of electrons to generate the metal atom. With a larger atomic mass, this meant a smaller n(electrons) was needed.

The second mark was awarded for showing an understanding that for the same period of time, a smaller n(electrons) needs to be transferred and hence a lower current is required to plate the same mass of rhodium.

Mathematically, marks were awarded for recognising:

- (first mark) to plate 7.8 g n(Rh) = 0.0758 mol therefore n($e^-$) = 0.2274 mol compared to 0.450 mol that was required for chromium.
- (second mark) over 2.5 hours of plating, 2.44 A is needed to plate the Rh, compared to the 4.83 A needed for chromium.

The most common misconceptions were:

- attempting to explain the difference in current needed by trying to compare voltages for the half equations
- failing to identify that the same charge was present on both cations
- failing to identify that current and time are inversely related to each other.

# Question 3a.

| Marks | 0 | 1 | Average |
|---|---|---|---|
| % | 12 | 88 | 0.9 |

Responses needed to state that a reversible reaction is one that can proceed in either direction.

No marks could be awarded if students attempted to use an equilibrium explanation, i.e. ‘rate of forward reaction = rate of reverse reaction’.

# Question 3b.i

| Marks | 0 | 1 | 2 | Average |
|---|---|---|---|---|
| % | 30 | 14 | 56 | 1.3 |

The first mark was awarded for a response indicating that increasing the temperature would cause a backward/reverse shift in the equilibrium based on the forward reaction being exothermic.

The second mark was awarded for then relating the backward/reverse shift to a decrease in yield.

# Question 3b.ii

| Marks | 0 | 1 | 2 | Average |
|---|---|---|---|---|
| % | 23 | 39 | 37 | 1.2 |

The first mark was awarded for stating a stress that would cause a forward shift, for example:

- removing the product
- adding a reactant
- increasing the pressure (but only if this was caused by decreasing the volume).

The second mark was awarded for providing the rationale as to how the stress caused the shift forward. System moved forward to partially:

- replace the loss of product
- remove the addition of reactant
- reduce the number of particles as LHS has 3 particles and RHS has 1 particle.

A significant number of responses just stated that an 'increase in pressure causes a forward shift'. This could not be awarded a mark, as an inert gas will increase the pressure but will not cause a shift forward. Hence responses were required to state not only 'increased pressure' but specifically the means by which it was achieved.

# Question 3c.

| Marks | 0 | 1 | 2 | 3 | Average |
|---|---|---|---|---|---|
| % | 16 | 19 | 53 | 12 | 1.6 |

The first mark was awarded for determining the value of the reaction quotient.

$$Q = 5.60 / (2.00 \times 1.25^3) = 1.792 \text{ M}^{-2}$$

The second mark was awarded for stating that Q > K.

The third mark was awarded for showing an understanding that to return Q to K, the system needed to shift backwards / reverse / move to the left so that the products are decreased and reactants increased.

# Question 3d.

| Marks | 0 | 1 | 2 | 3 | 4 | Average |
|---|---|---|---|---|---|---|
| % | 33 | 14 | 10 | 14 | 28 | 1.9 |

For any of these equilibrium type questions that involve an initial and final state, while these calculations can be done without the use of a table, it is highly recommended that an I.C.E. table is used, as shown below.

| | [ CO ] M | [ $H_2$ ] M | [ $CH_3OH$ ] M |
|---|---|---|---|
| Initial concentration | 5.00 | | 0.00 |
| Change | -x | -2x | +x |
| Final concentration | | | 1.30 |

From the table, x has to be equal to 1.30 M.

The first mark was awarded for establishing that [CO] = 5.00 – 1.30 = 3.70 M.

The second mark was awarded for using the equilibrium expression to find $[H_2]^2$.

$$[H_2]^2 = \frac{1.30}{3.70 \times 1.21} = 0.290 \text{ M}^2$$

The third mark was awarded for determining the $[H_2] = \sqrt{0.290} = 0.5394$ M.

The fourth mark was awarded for determining $n(H_2)_{initial}$.

$[H_2]_{initial} = 0.53954 + (2 \times 1.30 = 3.14$ M therefore $n(H_2) = 3.14$ mol

A significant number of students did not realise the need for completing both an I.C.E. table and also an equilibrium expression in order to complete this question.

For those that did complete the calculations, the most common mistake was that they left their final value as a concentration, not the amount in mol as specified in the stem of the question.

# Question 4a.i

| Marks | 0 | 1 | 2 | Average |
|---|---|---|---|---|
| % | 25 | 27 | 48 | 1.3 |

The first mark was awarded for the correct chemical equation.

The second mark was awarded for the inclusion of UV light as a requirement of the reaction.

$$CH_3CH_2CH_3 + Cl_2 \xrightarrow{\text{uv light}} CH_3CH_2CH_2Cl + HCl$$

# Question 4a.ii

| Marks | 0 | 1 | 2 | Average |
|---|---|---|---|---|
| % | 56 | 27 | 17 | 0.6 |

‘1-chloropropane’ and ‘ammonia’ were the two most common responses that were awarded these two marks. (Any primary haloalkane with 3 carbons was accepted.)

‘propan-1-ol’ was also accepted with ‘ammonia’.

The stem of the question stated that ‘names’ should be provided. However, a high proportion of students responded with chemical formulas.

# Question 4b.i

| Marks | 0 | 1 | 2 | Average |
|---|---|---|---|---|
| % | 35 | 37 | 29 | 1.0 |

The first mark was awarded for showing any of the following concepts:

- The amount of H-bonding between carboxylic acids is greater than the H-bonding between amines.
- Carboxylic acids can form dimers due to the nature of the COOH group.
- Carboxylic acids can form stronger intermolecular forces.

The second mark was awarded for showing an understanding that with either greater forces of attraction or greater masses, more energy is needed to separate the carboxylic acid molecules and hence results in higher boiling points.

Common errors included:

- discussion relating to the strength of C=O and O-H bonds
- stating that amines only have dispersion forces between molecules.

# Question 4b.ii

| Marks | 0 | 1 | Average |
|---|---|---|---|
| % | 23 | 77 | 0.8 |

amine

The compound could not be a carboxylic acid as there is no evidence for a carbonyl peak in the spectra or the broad O-H peak of an acid.

# Question 4c.i

| Marks | 0 | 1 | Average |
|---|---|---|---|
| % | 14 | 86 | 0.9 |

m/z = 59

The base peak is always shown to have a relative intensity of 100.

# Question 4c.ii

| Marks | 0 | 1 | Average |
|---|---|---|---|
| % | 59 | 41 | 0.4 |

The mark was awarded for recognising that all isomers will have the same molecular formulas and therefore all isomers will show the parent ion peak at m/z = 108.

Responses that stated that all isomers have the same molecular mass could not be awarded the mark as the molecular mass of the isomers is 108.5 g $mol^{-1}$.

An ideal response by a student would have looked something like the following.

'All isomers will have the $^{35}Cl$ isotope present in 75% of the molecules, and it is the presence of this isotope that is the only thing that produces the parent ion located at m/z = 108.'

# Question 4c.iii

| Marks | 0 | 1 | Average |
|---|---|---|---|
| % | 90 | 10 | 0.1 |

$[C_3H_6{}^{37}ClO]^+$

To be awarded this mark, students were required to provide a formula that clearly identified the $^{37}Cl$ isotope and had an overall single positive charge.

# Question 4d.

| Marks | 0 | 1 | 2 | 3 | Average |
|---|---|---|---|---|---|
| % | 46 | 32 | 17 | 5 | 0.8 |

Frequently, responses did not relate the data from the spectra or the table to support the feature of the molecule.

A mark was awarded for any of the following (maximum of 3 marks):

- There were three peaks seen in the spectra, therefore the molecule must have three hydrogen environments.
- Each of the three peaks in the spectra are singlets, therefore the molecule must have each hydrogen environment located with no adjacent hydrogens.
- The relative peak area of 16.7:33.3:100 shows that the ratio of hydrogens in the molecule is 1:2:6.
- The peak ratios of 1:2:6 show that there is likely to be a total number of 9 hydrogens in the molecule.
- The 6H's suggested from the peak area at 1.3 ppm suggests symmetry of two $-CH_3$ groups in the molecule.

Some students attempted to use chemical shifts to justify suggested structural features, but this could not be awarded any marks as the shifts are not reliable and cannot be used effectively in this manner.

## Question 4e.

| Marks | 0 | 1 | 2 | Average |
|---|---|---|---|---|
| % | 17 | 64 | 19 | 1.0 |

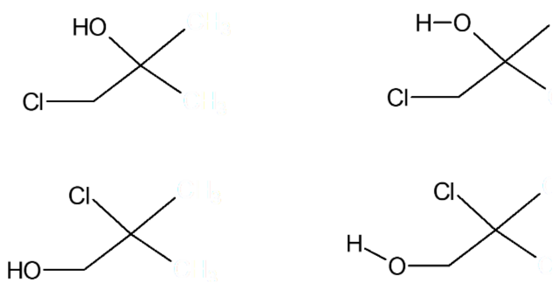
A mark was awarded for each of the following:

- The molecule must have three carbon environments.
- Symmetry must be present in the molecule in order for the four carbons to show only three peaks.

As in Question 4d., some students attempted to use chemical shifts to justify suggested structural features, but this could not be awarded any marks as the shifts are not reliable and cannot be used effectively in this manner.

## Question 4f.i

| Marks | 0 | 1 | Average |
|---|---|---|---|
| % | 85 | 15 | 0.2 |



Either isomer of $C_4H_9ClO$ was accepted but it had to be shown in its skeletal form.

## Question 4f.ii

| Marks | 0 | 1 | Average |
|---|---|---|---|
| % | 73 | 27 | 0.3 |

Either 1-chloro-2-methylpropan-1-ol or 2-chloro-2-methylpropan-1-ol.

## Question 5a.i

| Marks | 0 | 1 | 2 | Average |
|---|---|---|---|---|
| % | 52 | 27 | 20 | 0.7 |

The first mark was awarded for recognising that a quinine standard needed to be run under the same conditions and a retention time established.

The second mark was awarded for recognising that the retention time of the standard needed to match that of the peak in the tonic water in order for the presence of quinine to be possible.

Some students used the argument of ‘spiking’, which was also accepted.

The first mark was awarded for recognising that a sample of tonic water needed to be run and then a small quantity of pure quinine was then added to the sample.

The second mark was awarded for recognising that when the spiked sample was run again, for quinine to be identified as the peak at 1.415 mins, this peak must have increased in size in comparison to the first run.

# Question 5a.ii

| Marks | 0 | 1 | 2 | Average |
|---|---|---|---|---|
| % | 68 | 25 | 7 | 0.4 |

The first mark was awarded for recognising that uncontrolled conditions will produce peaks at various retention times and this has the potential to alter the validity of the measurement of peak areas.

The second mark was awarded for recognising that without valid peaks areas an inaccurate/invalid calibration curve will be generated.

# Question 5a.iii

| Marks | 0 | 1 | Average |
|---|---|---|---|
| % | 25 | 75 | 0.8 |

‘Cannot accurately extrapolate the data’ or ‘outside of the range of the standards’ were accepted.

# Question 5b.i

| Marks | 0 | 1 | Average |
|---|---|---|---|
| % | 43 | 57 | 0.6 |

Hydroxyl or hydroxy.

# Question 5b.ii

| Marks | 0 | 1 | 2 | Average |
|---|---|---|---|---|
| % | 21 | 33 | 46 | 1.3 |

The first mark was awarded for identifying Q as the correct structure.

The second mark was awarded for showing an understanding that there is a change in the functional groups at low pH, i.e. amino groups / amines / nitrogen / bases are subject to protonation / accepting $H^+$ ions.

# Question 5c.i

| Marks | 0 | 1 | 2 | 3 | Average |
|---|---|---|---|---|---|
| % | 31 | 28 | 27 | 14 | 1.3 |

The first mark was awarded for recognising that a chiral centre in a molecule arises due to four different substituents being present.

The second mark was awarded for recognising that these chiral centres generate uniquely shaped isomers.

The third mark was awarded for recognising that only one of these isomers will have the exact three-dimensional shape / complimentary shape needed to interact with the malaria enzyme’s active site.

# Question 5c.ii

| Marks | 0 | 1 | 2 | Average |
|---|---|---|---|---|
| % | 60 | 32 | 8 | 0.5 |

The first mark was awarded for recognising that if Q is a more effective inhibitor, it must bind more strongly to the active site.

The second mark was awarded for the chemical explanation of this by any of the following:

- The protonated form can form (ionic) bonds with negatively charged groups within the active site.
- The protonated form of Q has caused the shape of the molecule to change so it fits better in the active site.
- An inhibitor is a molecule that fits into the active site, blocking the substrate from entering.

# Question 6a.

| Marks | 0 | 1 | Average |
|---|---|---|---|
| % | 57 | 43 | 0.5 |

Literature review.

# Question 6b.

| Marks | 0 | 1 | Average |
|---|---|---|---|
| % | 78 | 22 | 0.2 |

The mark was awarded for recognising that during a permanganate titration, the end point is denoted when the solution starts to take on the colour of the titrant, i.e. first sign of permanent ‘purple’ colour.

‘Pink’ could not be accepted as Item 7 of the Databook states that the $Mn^{2+}$ ion is pink, and this will be steadily forming as the titration reaction takes place.

# Question 6c.i

| Marks | 0 | 1 | Average |
|---|---|---|---|
| % | 10 | 90 | 0.9 |

The mark was awarded for recognising that the increased temperature will cause a faster rate of reaction.

# Question 6c.ii

| Marks | 0 | 1 | Average |
|---|---|---|---|
| % | 34 | 66 | 0.7 |

Students responded to this question in two possible ways, both of which were accepted. Responses could be either:

- Room temperature would result in a reaction that was too slow to get valid results. OR
- Room temperature was not stated in the methodology, ‘boiling’ was required. Therefore, this would not be a valid step.

# Question 6d.

| Marks | 0 | 1 | Average |
|---|---|---|---|
| % | 35 | 65 | 0.7 |

Average titre = $\dfrac{15.45+15.40+15.40}{3}$ = 15.42 mL (0.01542 L)

# Question 6e.

| Marks | 0 | 1 | 2 | 3 | 4 | Average |
|---|---|---|---|---|---|---|
| % | 36 | 9 | 18 | 24 | 13 | 1.7 |

The first mark was awarded for the correct calculation of permanganate ion.

$n(MnO_4^-)_{titre}$ = 0.0020 x 15.42/1000 = 3.084 x $10^{-5}$ mol

The second mark was awarded for the correct calculation of oxalate ion in aliquot.

$n(C_2O_4^{2-})_{25\ mL}$ = 5/2 x 3.084 x $10^{-5}$ = 7.71 x $10^{-5}$ mol

The third mark was awarded for the correct calculation of oxalate ion in spinach sample.

$n(C_2O_4^{2-})_{500\ mL}$ = 500/25 x 7.71 x $10^{-5}$ = 1.542 x $10^{-3}$ mol (in 15.15 g of spinach)

The fourth mark was awarded for the correct calculation of mass of oxalate ion.

$m(C_2O_4^{2-})_{15.15\ g}$ = 88.0 x 1.542 x $10^{-3}$ = 0.136 g = 0.14 g (in 15.15 g of spinach)

The most common issue was students not allowing for dilution or reaction stoichiometry in their calculations.

# Question 6f.

| Marks | 0 | 1 | Average |
|---|---|---|---|
| % | 83 | 17 | 0.2 |

% reduction = $\dfrac{raw - cooked}{raw} \times 100 = \dfrac{0.136 - 0.0545}{0.136} \times 100$ = 60%

# Question 6g.

| Marks | 0 | 1 | 2 | Average |
|---|---|---|---|---|
| % | 61 | 34 | 5 | 0.5 |

The first mark was awarded for showing an understanding that residual water remaining on the outside of leaves would add soluble oxalate ions into the analysis.

The second mark was awarded for showing an understanding that this would lead to more permanganate being required or a larger titre of permanganate.

# Question 6h.

| Marks | 0 | 1 | 2 | Average |
|---|---|---|---|---|
| % | 44 | 20 | 36 | 0.9 |

The first mark was awarded for recognising that 75% was the predicted outcome in the hypothesis.

The second mark was awarded for making a comparison or a correct statement as to whether the result from Question 6f. was higher than, lower than or the same as the 75% predicted.

# Question 7a.

| Marks | 0 | 1 | 2 | Average |
|---|---|---|---|---|
| % | 53 | 32 | 15 | 0.7 |

Expected responses included:

| First mark | Second mark |
|---|---|
| The MOE process requires significantly more / a lot of electrical energy. | This electrical energy would need to come from renewable energy sources. |
| MOE produces no $CO_2$. | Therefore, it reduces the impact of steel production on global warming. |
| MOE does not require coal mining. | Which is a finite resource and/or results in significant land degradation. |
| MOE anode is made of:<br>• Pt which is rare and expensive<br>• Ir which is rare, expensive and radioactive. | Better to use the ceramic metal oxide and/or land degradation and/or finite resource. |

# Question 7b.

| Marks | | 0 | 1 | 2 | 3 | 4 | 5 | 6 | Average |
|---|---|---|---|---|---|---|---|---|---|
| % | | 11 | 5 | 11 | 15 | 20 | 22 | 15 | 3.6 |

The first mark was awarded for identifying one relevant green chemistry principle (GCP).

The second mark was awarded for using evidence to support/explain how the two methods relate to this GCP.

The third mark was awarded for identifying a second relevant GCP.

The fourth mark was awarded for using evidence to support/explain how the two methods relate to this GCP.

The fifth mark was awarded for providing a relevant ethical factor relating to global steel production.

The sixth mark was awarded for a concluding statement that related to the evidence the student provided.

**Examples that could have been used for green chemistry principles included:**

*Prevention of waste*

- Data to use
  - comparing the relative amounts of $CO_2$ produced, OR
  - comparing both $CO_2$ and $O_2$ as waste/by-product with respect to their environmental impact.
- The blast furnace process has a by-product of little industrial use ($CO_2$), so that becomes waste. MOE has no direct $CO_2$ emissions.
- The MOE process has a by-product ($O_2$), but it is less harmful and has some industrial use, so it is not as harmful a 'waste' by-product as $CO_2$.

*Design for energy efficiency*

- Data to use
  - comparing temperature requirements of the two processes, OR
  - comparing electrical energy requirements of the two processes.
- The blast furnace needs more heat energy (2200 compared to 1600). This is more likely done through the burning of fossil fuels.
- The MOE process needs more electrical energy than the blast furnace process. It could be argued that the former is not as energy efficient if done by using fossil fuels to produce the electricity. However, it could also be more energy efficient if both the heating stage and the electrolysis stage are done using electricity from renewable energy sources.

Note: Therefore, students could use data to justify either the blast furnace process OR the MOE process for this GCP.

*Atom economy*

- Data to use
  - atom economy of two reactions, with Fe as the desired product.
- Fe from BF process – 62.8%
- Fe from MOE process – 69.9%

If the worded explanation is reasonable, students would not have to provide detailed mathematical proofs. For example, the blast furnace process has *more* mass of reactants (C, coke) and an undesired by-product with *more* mass ($CO_2$ compared to $O_2$), so logically the atom economy of the MOE process should be better.

**Examples that could have been used for ethical factors included:**

Any discussion connecting the argument that the MOE process would only be green and sustainable if electricity from renewable sources is used, and that *access* to this will vary significantly across countries/globally.

For example:

- Renewable energy technology is not set up / accessible in all regions – some steel producers might want to change to the MOE process but cannot do so.
- Renewable energy is needed for a lot of things, so an ethical question arises about whether the steel industry should receive priority access over other people/communities/industries.
- Steel is produced on a massive scale and provides many jobs; a transition to the MOE process would require large numbers of workers to retrain, and some may lose jobs.
- The MOE process is relatively untested and will take many years to scale up. Meanwhile, society will continue to rely on steel production.
- Steel, as a polluting industry, is predominantly produced today in low- to middle-income countries or disadvantaged areas of industrial countries. There is potential for these areas to benefit from a process that uses renewable energy, including health benefits.
- Platinum and iridium are already difficult-to-source (endangered) elements; a global switch to the MOE process would significantly increase demand and extraction of these metals, with hard-to-predict environmental and social consequences.